

# MATH337 Reference

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## 1 Chapter 1

### 1.1 Traffic Intensity

$$I = \frac{La}{R}$$

where

- $L$  is the packet size [bits/packet],
- $a$  is the arrival rate [packets/sec],
- $R$  is the transmission rate [bits/sec].

Note: as traffic intensity approaches 1, the queuing delay approaches infinity asymptotically.

### 1.2 Transmission Delay

$$\begin{aligned} \text{Node-to-node: } d_{\text{trans}} &= \frac{L}{R} \\ \text{End-to-end: } d_{\text{trans}} &= N \frac{L}{R} \end{aligned}$$

where

- $L$  is the packet size [bits],
- $N$  is the number of nodes,
- $R$  is the transmission rate [bits/sec].

### 1.3 Propagation Delay

$$d_{\text{prop}} = \frac{d}{s}$$

where

- $d$  is the distance [meters],
- $s$  is the speed of light [meters/sec].

### 1.4 Queuing Delay

$$d_{\text{queue}} = \frac{I \cdot (L/R)}{1 - I}$$

where

- $I$  is the traffic intensity,
- $L$  is the packet size [bits],
- $R$  is the transmission rate [bits/sec].

### 1.5 Total Nodal Delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

where

- $d_{\text{proc}}$  is the processing delay,
- $d_{\text{queue}}$  is the queuing delay,
- $d_{\text{trans}}$  is the transmission delay,
- $d_{\text{prop}}$  is the propagation delay.

### 1.6 End-to-End Throughput

$$\min \{R_1, R_2, \dots, R_N\}$$

where  $R_i$  is the transmission rate at node  $i$ . Assuming that there is only one source and one destination, the end-to-end throughput is dependent on the bottleneck link's transmission rate.

## 2 Chapter 2

### 2.1 Client-Server File Distribution

$$D_{cs} \geq \max \left\{ \frac{NF}{u_s}, \frac{F}{d_{\min}} \right\}$$
$$d_{\min} = \min \{d_1, d_2, \dots, d_N\}$$

where

- $N$  is the number of peers,
- $F$  is the file size [bits],
- $u_s$  is the server upload rate [bits/sec],
- $d_{\min}$  is the minimum peer download rate [bits/sec],
- $d_i$  is the download rate of peer  $i$  [bits/sec].

This assumes one server is used to distribute the file to multiple clients. Oftentimes, the  $\geq$  can be replaced with  $=$ .

### 2.2 Peer-to-Peer File Distribution

$$D_{P2P} \geq \max \left\{ \frac{F}{u_s}, \frac{F}{d_{\min}}, \frac{NF}{u_s + \sum_{i=1}^N u_i} \right\}$$

where

- $N$  is the number of peers,
- $F$  is the file size [bits],
- $u_s$  is the server upload rate [bits/sec],
- $d_{\min}$  is the minimum peer download rate [bits/sec],
- $u_i$  is the upload rate of peer  $i$  [bits/sec].

This assumes that the server only needs to send one copy of the file to the peers, who can then redistribute it among themselves. Oftentimes, the  $\geq$  can be replaced with  $=$ . Note: client-server distribution time scales linearly with  $N$ , while peer-to-peer distribution time scales much slower with  $N$ , always remaining under the client-server distribution time.

## 3 Chapter 3

### 3.1 Utilization Rate

$$\text{Stop-and-wait: } U_{\text{sender}} = \frac{L/R}{RTT + L/R}$$

$$\text{Pipelined: } U_{\text{sender}} = \frac{NL/R}{RTT + L/R}$$

where

- $L$  is the packet size [bits],
- $R$  is the transmission rate [bits/sec],
- $N$  is the number of packets sent at once before receiving an ACK,
- $RTT$  is the round-trip time [sec].

Measures the fraction of time the sender is busy sending data.

### 3.2 Estimated RTT

$$\text{EstimatedRTT} = (1 - \alpha) \cdot \text{EstimatedRTT} + \alpha \cdot \text{SampleRTT}$$

where

- $\alpha$  is the smoothing factor (recommended  $\alpha = 0.125$ ),
- $\text{EstimatedRTT}$  is the estimated round-trip time [sec],
- $\text{SampleRTT}$  is the sample round-trip time [sec].

This uses an exponential weighted moving average (EWMA), in which the weight of a particular **SampleRTT** decreases exponentially fast with more updates.

### 3.3 RTT Deviation

$$\text{DevRTT} = (1 - \beta) \cdot \text{DevRTT} + \beta \cdot |\text{SampleRTT} - \text{EstimatedRTT}|$$

where

- $\beta$  is the smoothing factor (recommended  $\beta = 0.25$ ),
- $\text{DevRTT}$  is the round-trip time deviation [sec],

- SampleRTT is the sample round-trip time [sec],
- EstimatedRTT is the estimated round-trip time [sec].

This uses an EWMA on the absolute difference between the **SampleRTT** and **EstimatedRTT**.

### 3.4 Timeout Interval

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$$

where

- TimeoutInterval is in [sec] (initially 1 second),
- EstimatedRTT is the estimated round-trip time [sec],
- DevRTT is the round-trip time deviation [sec].

The timeout interval determines when to retransmit a packet if no ACK is received. After a timeout event, the timeout interval is doubled, but after successful receipt of an ACK, the timeout interval is again calculated using the formula above.

### 3.5 TCP Send Rate

$$\frac{\text{cwnd}}{\text{RTT}}$$

where

- cwnd is the congestion window [bytes],
- RTT is the round-trip time [sec].

### 3.6 Receive Window (Flow Control)

$$\begin{aligned} \text{rwnd} &= \text{RcvBuffer} - [\text{LastByteRcvd} - \text{LastByteRead}] \\ \text{RcvBuffer} &\geq \text{LastByteRcvd} - \text{LastByteRead} \end{aligned}$$

where

- rwnd is the receive window [bytes],
- RcvBuffer is the receive buffer [bytes],

- LastByteRcvd is the last byte in the receive buffer from the network layer [bytes],
- LastByteRead is the last byte read from the receive buffer by the application layer [bytes].

Informally, a measure of the spare room in the receive buffer.

### 3.7 TCP Unacknowledged Data Limit

$$\text{LastByteSent} - \text{LastByteAcked} \leq \min\{\text{cwnd}, \text{rwnd}\}$$

where

- LastByteSent is in [bytes],
- LastByteAcked is in [bytes],
- cwnd is the congestion window [bytes],
- rwnd is the receive window [bytes].

### 3.8 TCP Reno Average Throughput (Macroscopic Model)

$$\text{average throughput of a connection} = \frac{0.75 \cdot W}{\text{RTT}}$$

where

- average throughput of a connection is in [bits/sec],
- W is the maximum congestion window size, which occurs during a loss event [bytes],
- RTT is the round-trip time [sec].

### 3.9 TCP Average Rate with Loss

$$\approx \frac{1.22 \cdot \text{MSS}}{\text{RTT} \sqrt{L}}$$

where

- MSS is the maximum segment size [bytes],
- RTT is the round-trip time [sec],
- L is the loss rate.

## 4 Chapter 4

### 4.1 Buffer Size in Routers

Traditional wisdom suggests:

$$B = RTT \cdot C$$

However, more recent research suggests:

$$B = \frac{RTT \cdot C}{\sqrt{N}}$$

where

- $RTT$  is the round-trip time [sec],
- $C$  is the link capacity [bits/sec],
- $N$  is the *independent* TCP flows sharing the link.

Based on more recent research, this formula applies when  $N$  is large such as in the network core.

### 4.2 Weighted Fair Queuing (WFQ)

In output port packet scheduling, each class  $i$  is guaranteed to receive a fraction of the service, given by

$$\frac{w_i}{\sum_{j=1}^N w_j}$$

where

- $w_i$  is the weight of class  $i$ ,
- $N$  is the number of classes with at least one packet in the queue.

## 5 Chapter 5

### 5.1 Link-State Algorithm Time Complexity

Without a heap to choose the next node with the least cost, the time complexity is

$$O(n^2)$$

where  $n$  is the number of nodes.

## 5.2 Bellman-Ford Algorithm

$$d_x(y) = \min_v \{c(x, v) + d_v(y)\}$$
$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

where

- $c(x, v)$  is the cost of the edge from node  $x$  to node  $v$ ,
- $d_x(y)$  is the shortest path from node  $x$  to node  $y$ ,
- $D_x(y)$  is the estimation of the least cost from node  $x$  to node  $y$ ,
- $d_v(y)$  is the shortest path from node  $v$  to node  $y$ ,
- $D_v(y)$  is the estimation of the least cost from node  $v$  to node  $y$ ,
- $v$  is the set of neighbors of node  $x$ .

Eventually each  $D_x(y)$  will converge to  $d_x(y)$ .

## 5.3 Link-State Message Complexity

The number of messages that need to be sent is

$$O(|N||E|),$$

where

- $|N|$  is the number of nodes (routers),
- $|E|$  is the number of edges (links).

# 6 Chapter 6

## 6.1 Cyclic Redundancy Check (CRC)

$$R = \text{remainder} \left( \frac{D \cdot 2^r}{G} \right)$$

where

- $R$  is the remainder bits (length  $r$ ),
- $D$  is the data bits (length  $d$ ),
- $r$  is the number of remainder bits,
- $G$  is the generator of length  $r + 1$ .



The resulting bit string is  $D$  followed by  $R$  of total length  $d + r$ . Note:

- Can detect all burst errors of length  $r$  or less
- Can detect any odd number of errors
- Can detect burst errors of length  $r + 1$  with probability  $1 - 0.5^r$

## 6.2 Slotted ALOHA Efficiency

When there are a large number of nodes with many frames to trasmit, the maximum efficiency is:

$$\frac{1}{e} = 0.37$$

## 6.3 ALOHA Efficiency

When there are a small number of nodes with few frames to trasmit, the maximum efficiency is:

$$\frac{1}{2e} = 0.18$$

## 6.4 CSMA/CD Efficiency

When there are a large number of nodes with many frames to trasmit, the maximum efficiency is:

$$\frac{1}{1 + 5d_{\text{prop}}/d_{\text{trans}}}$$

where

- $d_{\text{prop}}$  is the maximum amount of time for a signal to propagate between any two adapters,
- $d_{\text{trans}}$  is the time to transmit a maximum-size frame.

# 7 Chapter 7

## 7.1 CDMA Encoding/Decoding

### 7.1.1 Single Sender

$$Z_{i,m} = d_i \cdot c_m \quad (\text{Encoding})$$

$$d_i = \frac{1}{M} \sum_{m=1}^M Z_{i,m} \cdot c_m \quad (\text{Decoding})$$

where

- $d_i$  is the  $i$ -th bit in the original signal,
- $Z_{i,m}$  is the encoded bit in the  $m$ -th mini-slot of  $d_i$ ,
- $c_m$  is the  $m$ -th bit of the assigned CDMA code,
- $M$  is the length of the code,
- $m = 1, 2, \dots, M$ .

### 7.1.2 Multiple Senders

$$Z_{i,m}^* = \sum_{s=1}^N Z_{i,m}^s \quad (\text{Encoding})$$

$$d_i = \frac{1}{M} \sum_{m=1}^M Z_{i,m}^* \cdot c_m \quad (\text{Decoding})$$

where

- $d_i$  is the  $i$ -th bit in the original signal,
- $Z_{i,m}^*$  is the encoded bit in the  $m$ -th mini-slot of  $d_i$  at the receiver,
- $Z_{i,m}^s$  is the encoded bit in the  $m$ -th mini-slot of  $d_i$  sent by sender  $s$  (computed exactly as in the single-sender case),
- $c_m$  is the  $m$ -th bit of the assigned CDMA code,
- $N$  is the number of senders,
- $M$  is the length of the code.

## 7.2 5G Capacity

$$\text{capacity} = \text{cell density} \times \text{available spectrum} \times \text{spectral efficiency}$$

where

- cell density is in  $[\text{cells}/\text{km}^2]$ ,
- available spectrum is in  $[\text{Hz}]$ ,
- spectral efficiency is how efficiently each base station can communicate with users in  $[\text{bps}/\text{Hz}/\text{cell}]$ .